A3: Building the Structure of a Website (HTML + Semantic HTML + Accessibility)

LIS 351 –Individual Assignment

Assignment Description

In this assignment, you'll plan and hand-code the HTML structure for a one-page website modeled after a professional landing page layout and includes a hero section, feature columns, content sections, and footer like the one below. Again, the goal is not to become a "web developer," but to learn how web information is structured so that browsers, search engines, and assistive technologies can interpret it. This assignment requires that you plan and then hand code the HTML that will structure the content on the page (Fig 1, left). You'll reuse the exact HTML file in the next assignment when you style it with CSS (Fig. 1, right).

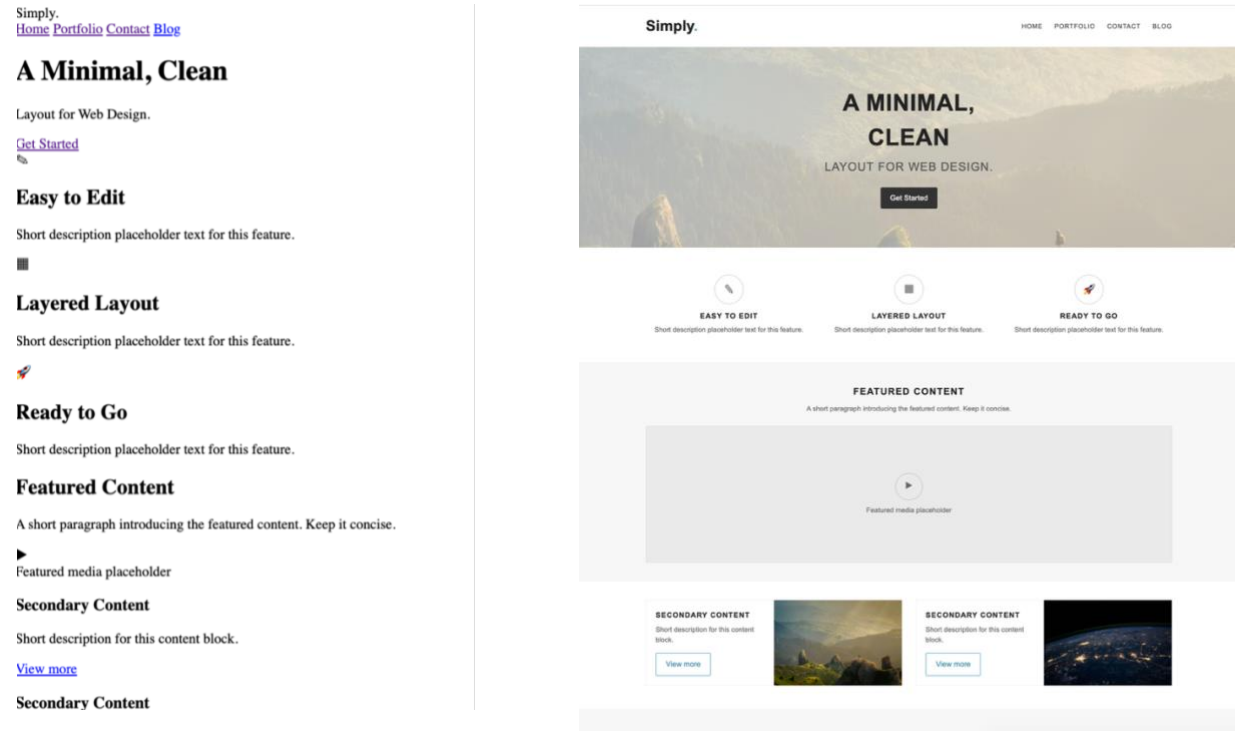


Fig 1. The web page with HTML for Weeks 2-3 (left) and the web page with CSS for Weeks 4-5 (right).

The learning objectives for this A3 are:

- Plan a webpage by outlining its purpose, audience, and content before coding
- Create a valid HTML5 document from scratch
- Build a page with multiple sections using headings, paragraphs, images, lists, and links
- Improve structure using semantic elements (`<header>`, `<nav>`, `<main>`, `<section>`, `<footer>`)
- Apply basic accessibility practices (alt text, descriptive link text, logical heading hierarchy)
- Validate HTML and correct common errors

Assignment Instructions

This assignment is broken into two parts – planning your website and coding with HTML. Follow the steps below to complete this assignment.

Part 1 (Week 2): Planning + Basic HTML Structure

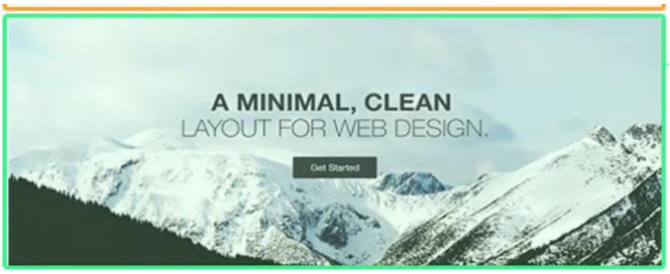
In this part, you will complete planning for your website. Before coding, you need to have a clear idea of what your page is about, who it is for, and what content it will include. Planning first will make the HTML much easier to write and revise later. Follow the steps below to organize your ideas and prepare the content you will use when coding.

Step 1 - Planning
Before completing this step it would be beneficial to read: What will your website look like? from this week's instructional content on Canvas. In this step, you **create a one-page plan** that includes:

1. Purpose: What is this page about? (topic/theme). You can choose to create a page about a topic of personal interest (e.g., a hobby, place, organization, cause, or project), a fictional company or product, or a simple informational page (e.g., "About Me," "Course Resources," or "Travel Guide"). The topic should give you enough content to fill all required sections of the page
2. Audience: Who do you think the primary audience for this web page will be? Why?
3. Content inventory: What text and images will you include? You should choose content that matches the purpose of your website. The table below shows where the content will be displayed once we apply the CSS. You should complete the following sections adding the text that will be displayed and the name of the image that will be used e.g., *mountans.jpeg*

Hero section (headline + short subhead)

Note: The hero image should be landscape-oriented with dimensions of at least 1200 × 600 pixels so it displays well across screen sizes. You can find free images with attribution at sites such as Unsplash or Pexels. Be sure to record where each image came from.



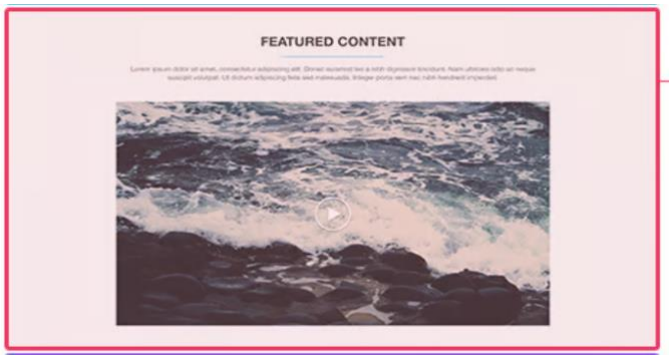
Three "feature" blurbs (short title + 1–2 sentences each)

Note: You can find simple icons by searching for "free SVG icons" or by using icon libraries such as Heroicons or Font Awesome (free icons only). Icons should be simple and visually consistent.

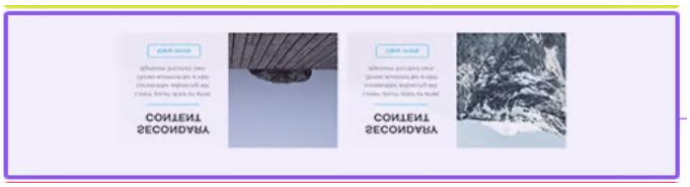


Featured content section (short paragraph + image/video placeholder).

Note: The featured content section should highlight something central to your page (for example, a key idea, project, location, or resource).



Two secondary content blocks (title + 1–2 sentences each)



Footer (basic links or text)



4. Color themes: What is the overall color theme of your webpage? A common approach is to select one primary color and a small set of complementary or analogous colors. One method is to choose a primary color using MDN's Color format converter, copy the RGB or HEX code, and then use Adobe's Color Wheel to explore color harmony options. Experiment with different combinations to determine the overall look and feel of your page. Once you select your colors, download the palette or include a screenshot showing each color with its HEX value (#7392FA).

5. Sketch: Create a simple sketch of your page layout (hand-drawn or digital). Even though everyone will use the same overall structure, sketching helps you think through where your content will appear and how sections relate to one another. This does not need to be detailed or artistic, boxes and labels are sufficient.

Step 2 - Folder structure
Create a project folder on your computer with the name ***lis351-web***/ and inside that folder another folder called ***images***/ (this is where you'll store any images you use for your website).

Step 3 - Build the basic HTML page
Now that you have all the content, you are ready to begin coding your web page. In your *lis351-web/* folder, create *index.html* with:

- A valid HTML5 structure: <!DOCTYPE html>, <html>, <head>, <body>
- A <title> in the head
- A top area that includes:
  - site name/logo text
  - a simple navigation menu with 3–4 links (they can be placeholders)
- A hero section with:
  - a large heading
  - a short sentence under it
  - a "Get Started" style button (link)
- A 3-column "features" section (3 feature blocks)
- A "featured content" section
- Two "secondary content" blocks
- A call-to-action section ("Heard enough?" + button)
- A footer with:
  - copyright line
  - at least one group of links

As you write HTML, be sure to open index.html in your browser and confirm it displays.

Part 2 (Week 3): Semantic HTML + Accessibility Revision

Now that the basic HTML has been completed, you should go back and revise your page so that its structure is meaningful, readable, and accessible. This step focuses on improving how your content is interpreted by browsers, search engines, and assistive technologies—not on visual styling.

Step 1 – Check semantic structure

Semantic HTML helps describe the role of different parts of your page. Instead of relying on generic containers, semantic elements communicate meaning (for example, navigation vs. main content). Using semantic HTML makes pages easier to understand, maintain, and access.Ensure your HTML includes:

- <header> for the top area
- <nav> for the menu
- <main> around your main content
- <section> for major content chunks
- <footer> for the bottom area

And that your headings follow a logical structure

- Exactly one <h1> (page title/headline)
- Section headings as <h2>
- Subheadings (if used) as <h3>

Step 2 - Accessibility checks:

Accessibility ensures that your webpage can be used by people with different abilities and by assistive technologies such as screen readers. Small changes in HTML can make a significant difference in usability and inclusion. Ensure:

- All images have **descriptive** alt **text**
- Links use descriptive text (avoid "click here")
- Sections are logically grouped and readable

Step 3 - Validate HTML:

Validating your HTML helps catch errors such as missing tags, improper nesting, or incorrect syntax. Valid HTML is more reliable across browsers and easier to maintain. Use an HTML validator (such as the W3C Validator) and fix any errors that appear. Once all validation errors are resolved, take a screenshot of the validation results and include it in your planning document.

What to Hand In

On Canvas, you should submit two documents:

Planning document (a PDF): The planning document should include:

Part I, *Step 1 – Planning* - text that will be displayed, images (in the \image folder) and image names, and URLs if you link other content
Part I, Step 2 – Folder Structure - screenshot of the folder with all sub folders opened so that I can view all images and index.html
Part II, Step 3 – Validate HTML - screenshot of the HTML validation page after errors have been fixed

The index.html file (final revised version after Part 2).

Grading Rubric

Criteria	Points	Description
Planning	2 pts	Your website has a clear purpose/audience + content plan + outline/sketch
Required sections present	2 pt	Your HTML file includes the following sections: nav, hero, 3 features, featured content, secondary blocks, CTA, footer
HTML Correctness	1 pts	valid structure, tags closed/nested, minimal/no validation errors
Content Requirements		Includes required elements: 2+ paragraphs, 1+ link, 1+ image, 1+ list; content is original and appropriate
Semantic Structure		use of header/nav/main/section/foote
Accessibility basics		alt text + descriptive links + readable heading structure